

Numerical Analysis Homework XXX

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I. Convert a decimal integer to a normalized FPN

Using Algorithm 4.8, calculate

$$477/2 = 238...1$$

$$238/2 = 119...0$$

$$119/2 = 59...1$$

$$59/2 = 29...1$$

$$29/2 = 14...1$$

$$14/2 = 7...0$$

$$7/2 = 3...1$$

$$3/2 = 1...1$$

$$1/2 = 0...1$$

$$\text{we know } 477 = (111011101)_2 = (1.11011101) \times 2^8$$

II. Convert a decimal fraction to a normalized FPN

Using Algorithm 4.8, calculate

$$0.6 \times 2 = 1.2, \text{ where the integer is 1 and the decimal is 0.2}$$

$$0.2 \times 2 = 0.4, \text{ where the integer is 0 and the decimal is 0.4}$$

$$0.4 \times 2 = 0.8, \text{ where the integer is 0 and the decimal is 0.8}$$

$$0.8 \times 2 = 1.6, \text{ where the integer is 1 and the decimal is 0.6}$$

$$0.6 \times 2 = 1.2, \text{ where the integer is 1 and the decimal is 0.2 (repeat)}$$

$$\text{Thus, } \frac{3}{5} = (0.10011001 \dots)_2 = (1.00110011 \dots)_2 \times 2^{-1}$$

III. Adjacent FPNs

Since x is a normalized FPN, it can be written as:

$$x = m \times \beta^e, \quad \text{where } m = 1.$$

The set of normalized FPNs in $\mathcal{F}$ with exponent e consists of numbers of the form:

$$m \times \beta^e, \quad \text{where } m = \frac{n}{\beta^{p-1}}, \quad n \in \{\beta^{p-1}, \beta^{p-1} + 1, \dots, \beta^p - 1\}.$$

The spacing between consecutive numbers in this exponent range is:

$$\Delta_e = \frac{1}{\beta^{p-1}} \cdot \beta^e = \beta^{e-p+1}.$$

Thus, the next number after x in exponent e is:

$$Z_R = \left(1 + \frac{1}{\beta^{p-1}}\right) \beta^e = \beta^e + \beta^{e-p+1}.$$

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Now, consider numbers with exponent $e - 1$. The normalized FPNs with exponent $e - 1$ are:

$$m \times \beta^{e-1}, \quad \text{with } m = \frac{n}{\beta^{p-1}}, \quad n \in \{\beta^{p-1}, \beta^{p-1} + 1, \dots, \beta^p - 1\}.$$

The largest number in this set is:

$$M_{\max} \times \beta^{e-1} = \frac{\beta^p - 1}{\beta^{p-1}} \cdot \beta^{e-1} = \left(\beta - \frac{1}{\beta^{p-1}} \right) \beta^{e-1} = \beta^e - \beta^{e-p}.$$

This number is less than x , and there is no normalized FPN between this number and x because any number with exponent $e - 1$ is at most this value, and any number with exponent e is at least x . Therefore,

$$Z_L = \beta^e - \beta^{e-p}.$$

Then we compute:

$$x - Z_L = \beta^e - (\beta^e - \beta^{e-p}) = \beta^{e-p},$$

and

$$Z_R - x = (\beta^e + \beta^{e-p+1}) - \beta^e = \beta^{e-p+1}.$$

Thus,

$$Z_R - x = \beta^{e-p+1} = \beta \cdot \beta^{e-p} = \beta(x - Z_L).$$

This completes the proof.

IV. Adjacent FPNs for $x = 3/5$ under IEEE 754 Single-Precision

Under the IEEE 754 single-precision protocol, the parameters are:

- Base $\beta = 2$
- Precision $p = 24$ (including the implicit leading bit)
- Exponent range: $L = -126$, $U = 127$

From II, we know $\frac{3}{5} = (1.001100110011\dots)_2 \times 2^{-1}$. Thus, the exponent is $e = -1$, and the exponent field is $-1 + 127 = 126$, represented in binary as 01111110.

The significand $s = 1.2$ in decimal, so $s - 1 = 0.2$. The integer value for the fraction part is computed as $k = \lfloor (s - 1) \times 2^{23} \rfloor = \lfloor 0.2 \times 8388608 \rfloor = \lfloor 1677721.6 \rfloor = 1677721$. Thus, the two adjacent normalized FPNs are:

$$Z_L = \left(1 + \frac{1677721}{2^{23}} \right) \times 2^{-1}$$

$$Z_R = \left(1 + \frac{1677722}{2^{23}} \right) \times 2^{-1}$$

In IEEE 754 binary format:

- Z_L : sign bit 0, exponent 01111110, fraction 00110011001100110011001
- Z_R : sign bit 0, exponent 01111110, fraction 00110011001100110011010

The value $\text{fl}(x)$ is the rounded value of x under the round-to-nearest mode. Since x is closer to Z_R than to Z_L , we have $\text{fl}(x) = Z_R$.

The absolute error is $|Z_R - x| = 0.4 \times 2^{-24}$. The relative roundoff error is:

$$\frac{|Z_R - x|}{|x|} = \frac{0.4 \times 2^{-24}}{0.6} = \frac{2}{3} \times 2^{-24} = \frac{2^{-23}}{3}$$

V.

VI.

VII.

VIII.